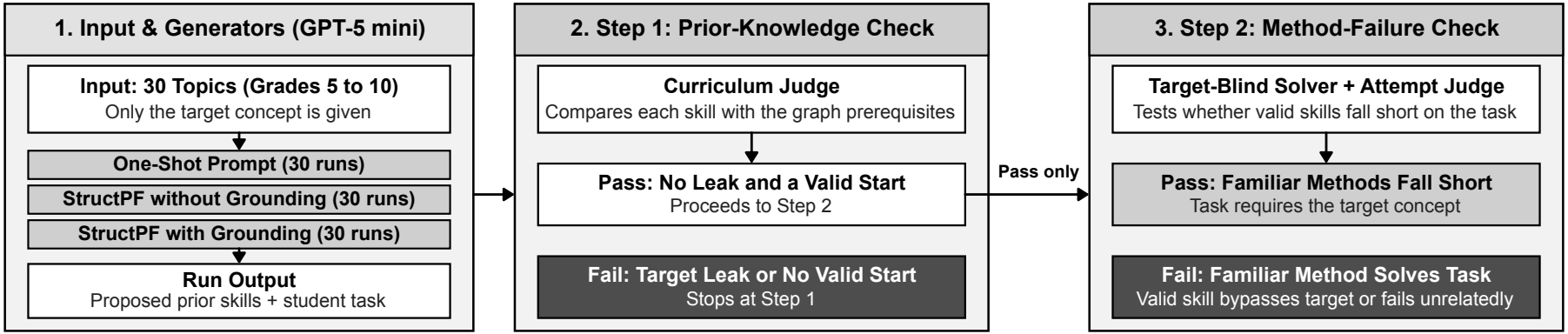
1. Input & Generators (GPT-5 mini)
Input: 30 Topics (Grades 5 to 10)
Only the target concept is given
One-Shot Prompt (30 runs)
StructPF without Grounding (30 runs)
StructPF with Grounding (30 runs)
Run Output
Proposed prior skills + student task
2. Step 1: Prior-Knowledge Check
Curriculum Judge
Compares each skill with the graph prerequisites
Pass: No Leak and a Valid Start
Proceeds to Step 2
Fail: Target Leak or No Valid Start
Stops at Step 1
Pass only
3. Step 2: Method-Failure Check
Target-Blind Solver + Attempt Judge
Tests whether valid skills fall short on the task
Pass: Familiar Methods Fall Short
Task requires the target concept
Fail: Familiar Method Solves Task
Valid skill bypasses target or fails unrelatedly